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Reporting Mediation and Moderation

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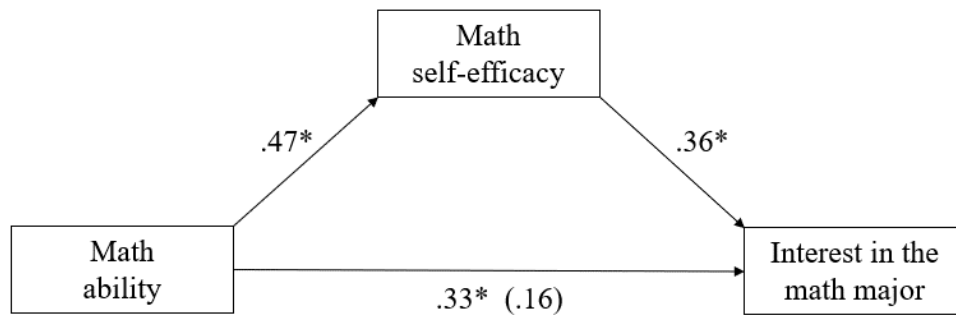
Complex regression procedures like mediation and moderation are best explained with a combination of plain language and a figure. For mediation, a path diagram that illustrates the mediational relationship and indicates beta weights is most useful. The statistical significance of the indirect effect should be tested using bootstrapping (see Hayes [2018], *Introduction to mediation, moderation, and conditional process analysis*). For moderation, a figure showing conditional/simple slopes at different levels of the moderator (typically 1 *SD* above, 1 *SD* below, and the *M* of the moderator variable for continuous moderators) is most useful.

A brief, simulated example of how to report simple mediation:

The relationship between math ability and interest in becoming a math major was mediated by math self-efficacy. As Figure 1 illustrates, the regression coefficient between math ability and math self-efficacy was statistically significant, as was the regression coefficient between math self-efficacy and interest in the math major. The indirect effect was $(.47)(.36) = .17$. We tested the significance of this indirect effect using bootstrapping procedures. Unstandardized indirect effects were computed for each of 10,000 bootstrapped samples, and the 95% confidence interval was computed by determining the indirect effects at the 2.5th and 97.5th percentiles. The bootstrapped unstandardized indirect effect was .17, and the 95% confidence interval ranged from .04 to .32. Thus, the indirect effect was statistically significant.

Figure 1

Regression Coefficients for the Relationship Between Math Ability and Interest in Becoming a Math Major as Mediated by Math Self-Efficacy



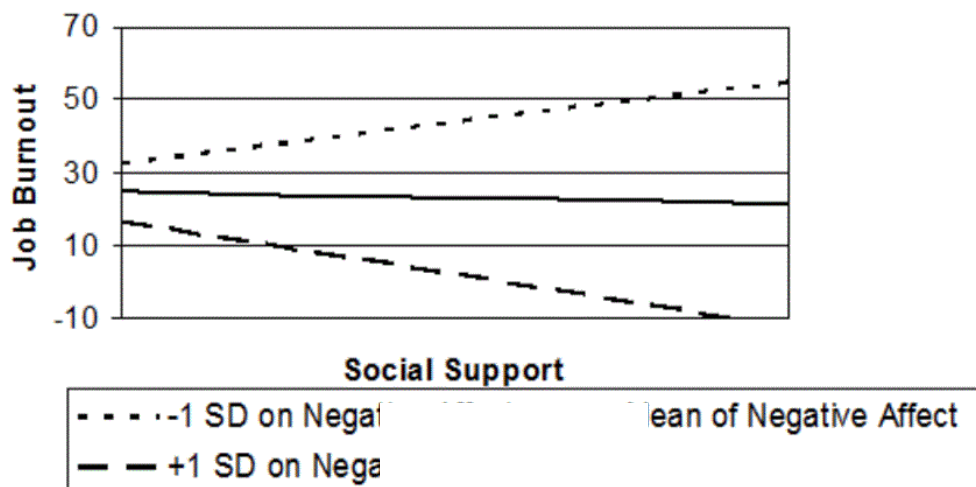
* $p < .05$.

A brief, simulated example of how to report moderation:

Negative affect was examined as a moderator of the relation between social support and job burnout. Social support and negative affect were entered in the first step of the regression analysis. In the second step of the regression analysis, the interaction term between negative affect and social support was entered, and it explained a significant increase in variance in job burnout, $\Delta R^2 = .03$, $F(1, 335) = 14.61$, $p < .001$. Thus, negative affect was a significant moderator of the relationship between social support and job burnout. The unstandardized simple slope for employees 1 SD below the mean of negative affect was .56, the unstandardized simple slope for employees with a mean level of negative affect was -.08, and the unstandardized simple slope for employees 1 SD above the mean of negative affect was -.72 (see Figure 2).

Figure 2

Simple Slopes of Social Support Predicting Job Burnout for 1 SD Below the Mean of Negative Affect, the Mean of Negative Affect, and 1 SD Above the Mean of Negative Affect





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